Exchange Student Guide

A moderated, community-written knowledge base for exchange students at IIT Madras

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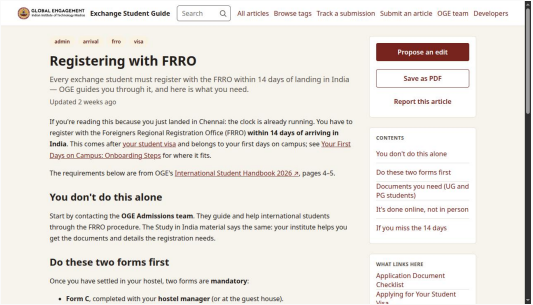
CS5013 Programming with AI · Mid-evaluation · Stakeholder: OGE, IIT Madras

The problem, and who uses it

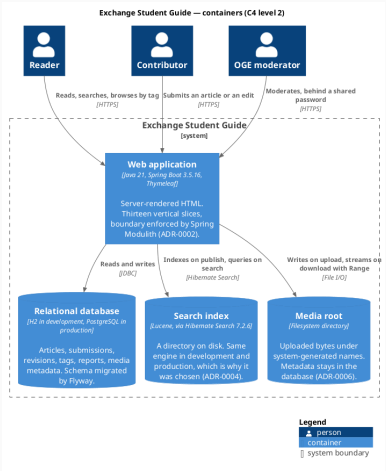
The information exists, but it is scattered across the IITM site, WhatsApp groups and blogs; a student three days off the plane cannot find it.

- Student reads: search, articles linked with `[[Title]]`, tags
- Student writes, with no account: a new article or an edit, with a photo
- OGE moderates: nothing is public until a moderator approves it

39 starter articles from OGE's Handbook 2026.

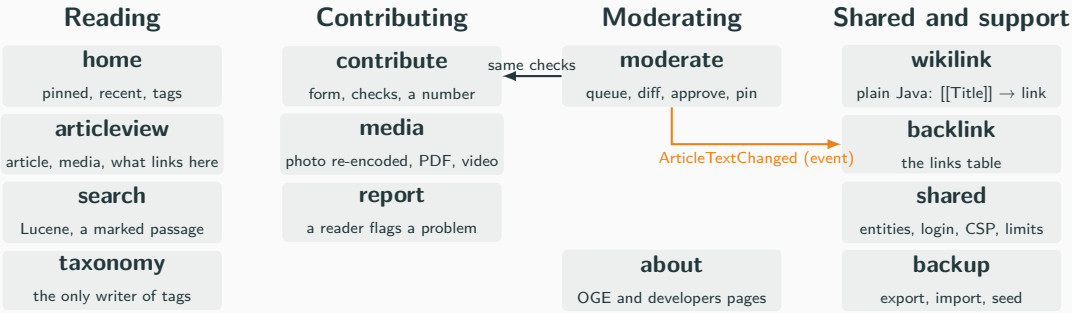


Architecture: one process, three stores



- Spring Boot 3.5, Java 21, server-rendered Thymeleaf — a modular monolith, not microservices: one team of two, one server at OGE
- PostgreSQL with Flyway: articles, every earlier version, submissions, tags, reports
- Lucene through Hibernate Search: same engine in tests and production; rebuilt at every start
- Media root: uploaded bytes on disk, metadata in the database
- Runs with `docker compose up`

Thirteen slices, and a boundary the build enforces



Inside every slice: root = published types · `web/` controllers · `internal/` services · `persistence/` repositories.

Another slice may use only the root, or an event; `ModularityTest` (Spring Modulith) and ArchUnit fail the build otherwise.

Data flow: an edit, from the student to the live article

1. contribute	form → rate limit (5/hour) → `SubmissionService.check` → a `Submission` row, tags, photo re-encoded → number `SUB-…`; the article is unchanged
2. moderate	queue → diff of the text as it reads (`TextDiff`) → approve or reject with a reason
3. approve	old text kept as a `Revision` → `Article` updated → media moved → event `ArticleTextChanged`
4. after commit	`backlink` re-reads the links; Hibernate Search updates the index — the change is live and searchable

Every request has the same shape: route → `web/` controller → `internal/` service → `persistence/` repository → Thymeleaf template.

What checks it

- 649 application tests, 356 for the tooling — every push
- The same suite on PostgreSQL 17 in CI
- 151 browser tests in Chromium with axe (accessibility), 320–1920 px
- Mutation testing: 130 of 132 mutants killed
- Property-based tests: 1000 titles in English, Hindi, Tamil

- A test is red before the code exists
- Test names are sentences: `ten_tags_are_accepted_and_eleven_are_refused`
- The demo rehearsal on Docker found 4 faults no unit test could see — each fixed with the test that catches it

Written with AI, and kept honest

- A contract before every change: goal, boundaries, acceptance criteria, open questions — the assistant waits for "yes". The criteria become the test names.
- Every prompt is in Git: hooks record 1126 prompts with outcomes and our own edits in `docs/ai/journal/`, committed automatically.
- Gates, not promises: a commit message without its requirement, a `TODO` without a debt entry, a stale generated document — refused at commit or push.
- Traceability: `//trace:FR-XXX` anchors, so requirement → code → test is generated, not written.
- Decisions written down: 22 ADRs, each with the options weighed.

Where we are

Done

- 42 of 43 requirements; the last, completing a wiki link while typing, is a *should*
- The main flow runs end to end on a reset stand
- Handoff for OGE: install behind HTTPS, moderator's guide, backup and restore checked

Open, on purpose

- 5 technical debt entries, each with its trigger — the gap list is generated, so it cannot be trimmed
- Next: OGE uses it on the stand; final demo and viva on 6 November

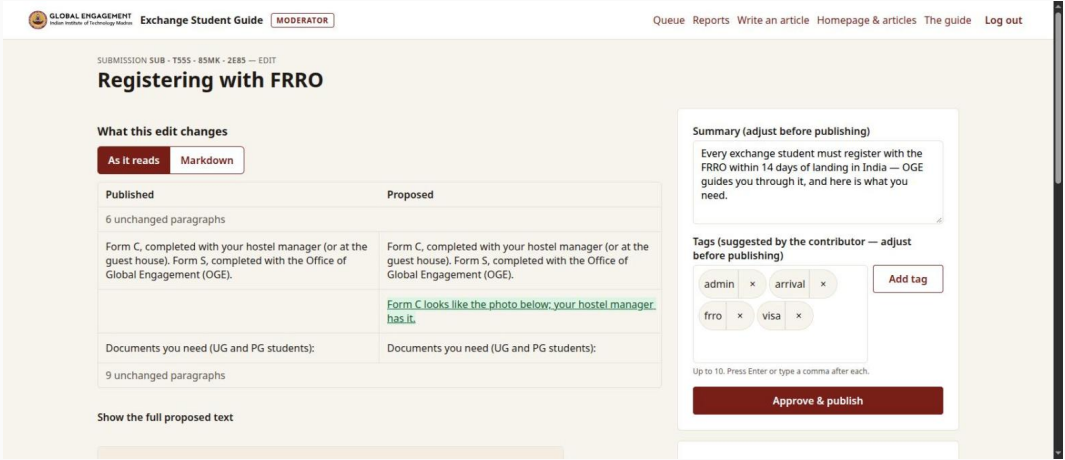
Questions?

github.com/rikire/Exchange-Student-Guide-Platform

Backup: where a typical change goes

Asked to…	Change, and check
change a limit	`application.yml` under `guide:`; tags `Tags.MOST`, summary `Article.LONGEST_SUMMARY` — the test of that limit
change wording on a page	`templates/<slice>/`; header and footer in `shared/web/Layout.html` — the controller's test
add a rule to the form	`SubmissionService.check` throws `SubmissionRejectedException(Field)` — a red case in `FieldErrorTest`
show a new field on an article	`ArticleController` model → `Article.html` — `ArticleControllerTest`
add a column	migration `V9__….sql`, entity in `shared/persistence/` — `SchemaMigrationTest`
add a route	controller, then `routes.yml` — `RouteContractTest`
change a colour	a token in `static/css/tokens.css` — `TemplateTokensTest`

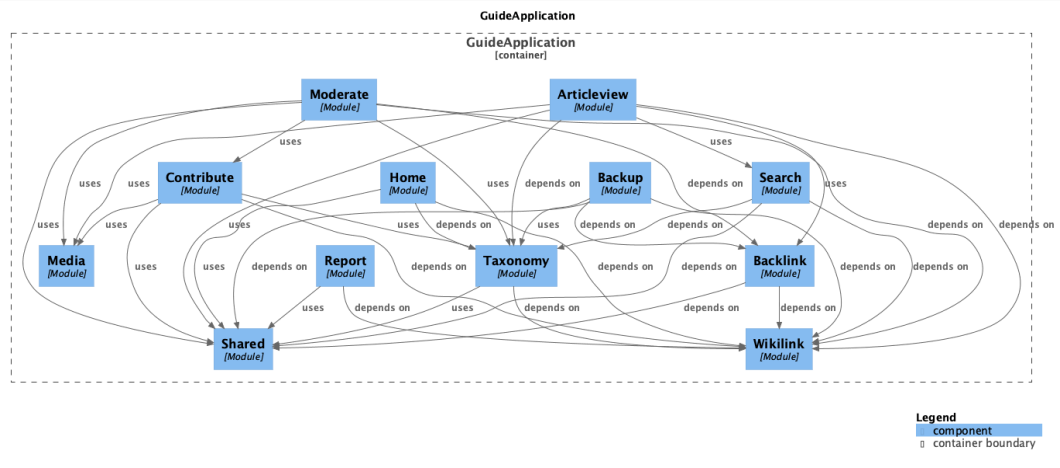
Backup: the moderator's review of an edit



Backup: decisions and the options we weighed

ADR-0002	Vertical slices in one process; Spring Modulith enforces the boundary
ADR-0003	A submission is its own table: an unapproved text is never a row of `article`; a revision is a full copy
ADR-0004	Search through Lucene, not the database: same engine in H2 tests and PostgreSQL
ADR-0009	One shared password for OGE, not accounts — students need none at all
ADR-0011	The submission number is an unguessable token, not a sequence
ADR-0016	One slice owns the links table, fed by an event in the writer's transaction
ADR-0022	Tag and file fields from libraries, and the CSP they need

Backup: the code itself draws its slices



Generated by Spring Modulith from the imports, in `ModularityTest`.